A per-subhalo convergence threshold at fixed $\psi_{\min}$: analytic structure of σ_κ

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1 What is being computed

The subhalo population of a host is held *fixed* at the production floor $\psi_{\min} = m_{\text{floor}}/M$ — every subhalo is always sampled. A separate knob $\kappa_{\text{thr,sub}}$ then acts on each clump's *actual convergence at the ray*,

$$\kappa_c(m, d), \qquad d = |\mathbf{y} - \mathbf{R}_{\text{clump}}| \text{ (lens-plane separation)}, \tag{1}$$

where $\mathbf{y}$ is the *ray impact parameter* — the lens-plane offset of the line of sight from the host centre — and $\mathbf{R}_{\text{clump}}$ is the projected clump position. Two distances must be kept apart: y measures ray-to-host-centre, d measures ray-to-clump. The threshold acts on d, but the distribution of d depends on y through $f(d\,|\,y)$, which is what couples them. Every moment below is conditional on y; the quoted numbers are area-weighted over the aperture,

$$\langle X \rangle = \int_0^{r_{\max}} X(y)\, q(y)\, dy, \qquad q(y) = \frac{2y}{r_{\max}^2}, \tag{2}$$

i.e. uniform ray density across the disc of radius $r_{\max}$, the radius at which the host's own convergence falls to the counting threshold κ_{thr} ($r_{\max} = 6705\,\text{kpc}$ at the fiducial host). dropping the clump from the sum whenever $\kappa_c < \kappa_{\text{thr,sub}}$. This is the same rule the *host* halos already obey: the `r_thr` table in `cpp/subhalo.cpp` is precisely "keep the object while its convergence at the ray exceeds κ_{thr}". The two thresholds are therefore directly comparable, which a $\psi_{\min}$ (mass-floor) sweep is not.

The cut does *not* commute with a mass cut: a massive clump far from the ray is dropped while a light clump the ray nearly hits survives. It must therefore be applied inside the clump–ray distance integral, not as a mass mask outside it.

Dropped clumps keep their mass in the smooth host (mass-conserving carve, scheme A), so the host is built at $M - M_{\text{ret}}$ where M_{ret} is the *retained* clump mass.

2 Campbell moments with a threshold

Clumps form a Poisson process with intensity $dN/d\ln\psi = \gamma\, \psi^\alpha e^{-\beta\psi^\omega}$ ($\alpha = -0.82$, $\beta = 50$, $\omega = 4$; $\psi = m/M$), and the clump–ray separation has the projected pdf $f(d\,|\,y)$ at ray impact parameter y. For a Poisson sum $S = \sum_i g(x_i)$ Campbell's theorem gives $\langle S \rangle = \int \lambda g$ and $\mathrm{Var}(S) = \int \lambda g^2$. A threshold only *restricts the domain*, so every moment stays an exact two-dimensional quadrature with the indicator $\Theta = \mathbf{1}[\kappa_c(m,d) \geq \kappa_{\rm thr,sub}]$ folded in:

$$\langle N_{\rm ret}|y\rangle = \int d\ln\psi\, \tfrac{dN}{d\ln\psi}\, P_{\rm ret}, \qquad \langle \kappa_{\rm sub}|y\rangle = \int d\ln\psi\, \tfrac{dN}{d\ln\psi}\, \langle \kappa_c\Theta\rangle_d, \tag{3}$$

$$\mathrm{Var}(\kappa_{\rm sub}|y) = \int d\ln\psi\, \tfrac{dN}{d\ln\psi}\, \langle \kappa_c^2\Theta\rangle_d, \qquad \langle M_{\rm ret}|y\rangle = \int d\ln\psi\, \tfrac{dN}{d\ln\psi}\, m\, P_{\rm ret}, \tag{4}$$

$$\mathrm{Var}(M_{\rm ret}|y) = \int d\ln\psi\, \tfrac{dN}{d\ln\psi}\, m^2\, P_{\rm ret}, \qquad \mathrm{Cov}(M_{\rm ret},\kappa_{\rm sub}|y) = \int d\ln\psi\, \tfrac{dN}{d\ln\psi}\, m\, \langle \kappa_c\Theta\rangle_d, \tag{5}$$

with $\langle X\rangle_d \equiv \int dd\, f(d|y)\, X$ and $P_{\rm ret}$ defined in Eq. (7). **The mean carries m, the variance carries m^2.** That single difference drives everything below.

3 The host is a linear response

The host convergence is $\kappa_{\rm host} = \kappa_{\rm NFW}(M - M_{\rm ret},\, y)$. Expanding about the mean carve,

$$\boxed{\sigma_{\rm host}(y) = \left|\frac{\partial \kappa_{\rm NFW}}{\partial M}\right| \sqrt{\mathrm{Var}(M_{\rm ret}|y)}} \tag{6}$$

Two consequences worth stating explicitly.

1. $M_{\rm ret}$ enters *only through its variance.* A deterministic carve, however large, gives $\sigma_{\rm host} = 0$: the host acquires scatter solely because $\sum_i m_i$ is redrawn in every realization. The host mass and the ray position are fixed; the realized subhalo mass is the only stochastic ingredient.

2. The *mean* carve $\langle M_{\rm ret}\rangle$ shifts $\langle \kappa_{\rm host}\rangle$ but touches $\sigma_{\rm host}$ only weakly, through the prefactor $\partial\kappa/\partial M$ evaluated at the reduced mass.

Note also that the carve is *not* spatial: the host is rebuilt as a fresh NFW of mass $M - M_{\rm ret}$ with r_s, ρ_s from the $c(M,z)$ relation, rather than hollowed out at the clump positions. Since the subhalo radial distribution is anti-biased, a spatially faithful carve would remove mass preferentially from the outskirts and change $\kappa_{\rm host}$ at small y much less. This is a known approximation of scheme A.

4 Where the threshold enters, and why $\sigma_{\rm host}$ is monotonic

$\kappa_c(m,d)$ decreases monotonically in d, so the cut is equivalent to a *reach radius*,

$$\kappa_c(m,d) \geq \kappa_{\rm thr,sub} \iff d \leq d_{\max}(m,\kappa_{\rm thr,sub}), \qquad P_{\rm ret}(m,y) = \int_0^{d_{\max}(m,\kappa_{\rm thr,sub})} f(d\,|\,y)\, dd. \tag{7}$$

The threshold appears in *one place only*, the upper limit of Eq. (7). Lowering $\kappa_{\rm thr,sub}$ increases $d_{\max}$ for *every* mass, so $P_{\rm ret}$ rises pointwise, so the integrand of $\mathrm{Var}(M_{\rm ret})$ rises pointwise. Hence

$$\frac{\partial\, \sigma_{\rm host}}{\partial\, \kappa_{\rm thr,sub}} \leq 0 \tag{8}$$

with no numerics required. The limits follow at once:

- $\kappa_{\rm thr,sub} \to \infty$: $d_{\max} \to 0$, $P_{\rm ret} \to 0$, $\mathrm{Var}(M_{\rm ret}) \to 0$, so $\sigma_{\rm host} \to 0$. Nothing is carved, the host is exactly M every realization, and it has no scatter.

- $\kappa_{\rm thr,sub} \to 0$: $P_{\rm ret} \to 1$ and the variance saturates at the finite constant

$$\mathrm{Var}_\infty = \gamma M^2 \int \psi^{\alpha+1} e^{-\beta\psi^\omega}\, d\psi, \tag{9}$$

convergent because $\alpha + 1 = 0.18 > -1$ at the low end and the exponential cuts off the high end.

Why the variance is top-heavy. In $d\ln\psi$ the integrands scale as

$$\text{mean: } \psi^{\alpha+1} = \psi^{0.18}, \qquad \text{variance: } \psi^{\alpha+2} = \psi^{1.18}. \tag{10}$$

Both rise toward large ψ, the variance far more steeply, so the carve *scatter* is set by the few heaviest subhalos while the carve *mean* collects the whole swarm. This is why the mean carved mass can double while $\sigma_{\rm host}$ moves by only a few percent.

Why saturation is slow. At large separation the projected NFW falls roughly as $\kappa_c \propto d^{-2}$, so $d_{\rm max} \propto \kappa_{\rm thr,sub}^{-1/2}$: retention grows only as a power law, and $P_{\rm ret} \to 1$ only once $d_{\rm max}$ exceeds the host's extent ($\sim r_{200} + r_{\rm max} \approx 7500\,\mathrm{kpc}$ at the fiducial host). Hence $\sigma_{\rm host}$ is still creeping upward at $\kappa_{\rm thr,sub} = 10^{-6}$ rather than being flat.

5 Decomposition of the total

With the carve, host and subhalos are anti-correlated:

$$\mathrm{Cov}(\kappa_{\rm host}, \kappa_{\rm sub}) = -\frac{\partial\kappa_{\rm NFW}}{\partial M}\,\mathrm{Cov}(M_{\rm ret}, \kappa_{\rm sub}) < 0, \tag{11}$$

since $\partial\kappa/\partial M > 0$ and $\mathrm{Cov}(M_{\rm ret}, \kappa_{\rm sub}) > 0$ (clump mass and clump convergence rise together). A realization drawing a heavier clump population has more $\kappa_{\rm sub}$ but a lighter host, hence less $\kappa_{\rm host}$. Therefore

$$\mathrm{Var}_{\rm tot} = \mathrm{Var}_{\rm host} + \mathrm{Var}_{\rm sub} + 2\,\mathrm{Cov}(\kappa_{\rm host}, \kappa_{\rm sub}) \tag{12}$$

can—and does—fall *below* $\mathrm{Var}_{\rm sub}$. This ordering is a prediction of mass conservation, not a numerical artifact: switch the carve off, leave the host at full M, and the covariance vanishes, at which point $\sigma_{\rm tot} > \sigma_{\rm sub}$ as naive addition suggests.

6 Numerical results

Fiducial host $M = 10^{14}\,M_\odot$, $z_l = 0.5$, $z_s = 1$, $\psi_{\rm min} = 10^{-7}$ fixed, area-weighted over the aperture. Host counting threshold $\kappa_{\rm thr} = 1.276 \times 10^{-4}$.

Component budget at $\kappa_{\rm thr,sub} = 0$, in units of $\mathrm{Var}_{\rm sub}$: $1 + 0.097 - 0.182 = 0.914$, i.e. $\sigma_{\rm tot}/\sigma_{\rm sub} = 0.956$, correlation coefficient $r = -0.29$ (deepening to -0.66 by $\kappa_{\rm thr,sub} \simeq 0.3$, the single-dominant-clump limit).

Isolating the two effects in Eq. (6) over $0 \le \kappa_{\rm thr,sub} \le 1.2 \times 10^{-2}$: freezing the host mass and letting only $\mathrm{Var}(M_{\rm ret})$ vary reproduces a factor 2.8 of the total factor 3.1 change in $\sigma_{\rm host}$ ($\simeq 92\%$); freezing $\mathrm{Var}(M_{\rm ret})$ and letting only the host mass vary gives 8%.

$\kappa_{\mathrm{thr,sub}}$	$\langle N_{\mathrm{ret}}\rangle$	mass carved	σ_{tot}	$\sigma_{\mathrm{tot}}/\sigma_{\mathrm{tot}}^{(0)}$
0 (all clumps)	3.38×10^{4}	21.4%	1.0017×10^{-3}	1.0000
1.34×10^{-6} ($\simeq$ model-3 default)	8.54	17.1%	1.0011×10^{-3}	0.9994
1.24×10^{-4} ($\simeq$ host κ_{thr})	0.146	10.3%	9.997×10^{-4}	0.9980
1.20×10^{-3}	1.28×10^{-2}	5.2%	9.908×10^{-4}	0.9891
8.39×10^{-3}	1.20×10^{-3}	1.0%	9.420×10^{-4}	0.9404
0.155	3.84×10^{-6}	0.006%	4.128×10^{-4}	0.4121

7 Caveats

- Equation (6) is a *first-order* response. The carve fluctuation is not small and not Gaussian: $\mathrm{sd}(M_{\mathrm{ret}})/M = 0.110$ with skewness 1.83 at $\kappa_{\mathrm{thr,sub}} = 0$, rising to skewness ~ 6 at the host threshold, where the carve is essentially "was one massive clump retained or not". σ_{host} is thus the least trustworthy of the three curves; σ_{sub} is exact Campbell with no linearization and dominates the total.

- σ_{host} and σ_{sub} are not separately physical. The threshold defines where the split between "explicitly realized clump" and "smooth host" falls, and moving it shuffles mass across that line, so both components must move. Only the total is meaningful, and the total is flat to 0.2% at the host threshold.

- All scatter here is substructure-induced: a smooth deterministic halo at fixed y has none, which is why $\sigma_{\mathrm{tot}} \to 0$ at large $\kappa_{\mathrm{thr,sub}}$. This is the single-host substructure contribution, not σ_κ of the universe.

- Certified for σ_κ and the mean only. Since the tail of $P(\ln \mu)$ is built from rare close encounters, and a κ threshold is precisely a cut on encounter closeness, a JSD check on $P(\ln \mu)$ is required before adopting the threshold in production.

- Realizing the computational saving needs a sampler restructure. Testing $\kappa_c(m, d) \geq \kappa_{\mathrm{thr,sub}}$ per clump requires m and d, so draw-and-reject would still instantiate all 3.4×10^4 clumps. The saving requires sampling from the *restricted* intensity: Poisson($\langle N_{\mathrm{ret}}\rangle(y)$) clumps drawn from the retention-conditioned distribution, never instantiating the rejects.

- Numerics: grids converged to $\sim 10^{-4}$ in n_d and n_{mass}; the baseline agrees with the independent ψ_{min} engine to 1.4×10^{-5}. The n_y aperture grid shifts all curves by a uniform 1.83%, which cancels in ratios.

Code

- Engine (exact Campbell quadrature): `playground/analytic/sigma_vs_subkappathr.py`
- Figure: `paper_prod/scripts/plot_fig_subhalo_sigma_vs_subkappathr.py` → `fig_subhalo_sigma_vs_sub`
- ψ_{min} companion (Fig. 5 machinery): `playground/analytic/plot_sigma_vs_psimin.py`